

Multiple Choice (10 marks)

1. When $(x\sqrt{x^3})^4$ is simplified, what is the exponent of x ?
 - (a) 5
 - (b) 10
 - (c) 2
 - (d) 12
2. Calculate $\left(\frac{1}{a}\right)^4 \cdot 2 \cdot a^4 + a^{2+1-3}$ when $a = 42$.
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
3. Compute $\binom{85}{82}$.
 - (a) 252
 - (b) 4680
 - (c) 98770
 - (d) 101170
4. Six 6-sided dice are rolled. What is the probability that three of the dice show prime numbers and the rest show composite numbers?
 - (a) $\frac{125}{2916}$
 - (b) $\frac{5}{16}$
 - (c) $\frac{5}{54}$
 - (d) $\frac{5}{55}$
5. A number is rounded to the nearest hundred. The result is 500. Which number could not be the number before it was rounded to the nearest hundred?
 - (a) 458
 - (b) 463
 - (c) 547
 - (d) 559

True / False (10 marks)

Answer True or False

6. True or False: The number 179,912 in word form is written as one hundred seventy-nine thousand twelve.
7. True or False: The number that makes the equation $2 \text{ over } 9$ equal to $14 \text{ over } ?$ is 15.
8. True or False: The correct answer to 'Find $(2^{20} + 2^{20} + 2^{20} + 2^{21}) \div 2^{17}$.' is '10'.
9. True or False: The expression $-\frac{1}{-3} \cdot \frac{1}{\{\sim \frac{1}{-3} \sim\}}$ simplifies to 1.
10. True or False: Five inches on the map represents 125 miles in reality, based on the scale of 1 inch equals 25 miles.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. How many numbers are in the following list: $-4, -1, 2, 5, \dots, 32$
12. Discuss the concept of vertical axis of symmetry in vowels. Identify which vowel lacks this symmetry and explain your reasoning with examples.

End of Paper